Unit 9: Corporate Finance

Introduction to Credit Risk

Credit risk is one of the most important risks faced by banks, financial institutions, and lenders. It refers to the possibility that a borrower may fail to repay the principal amount or interest on a loan according to the agreed terms. In simple words, whenever a bank lends money to an individual or business, there is always uncertainty regarding whether the borrower will repay the amount on time. If the borrower fails to do so, the lender suffers financial loss. This possibility of loss is called credit risk.

Credit risk exists in various financial activities such as personal loans, corporate loans, credit cards, mortgages, bonds, and trade credit. For example, when a bank gives a housing loan, it expects regular EMI payments. If the borrower loses employment or faces financial difficulties and stops repayment, the bank faces default risk. Therefore, understanding and measuring credit risk is crucial for financial stability.

In modern finance, institutions do not rely only on intuition while lending. Instead, they use scientific methods known as credit risk modelling to estimate and manage borrower risk.

Credit Risk Modelling: Concept and Importance

Credit risk modelling is the process of using quantitative techniques, statistical tools, historical borrower data, and financial analysis to estimate the likelihood of default and the possible loss arising from that default. It helps financial institutions make informed lending decisions by systematically evaluating the borrower's ability and willingness to repay debt.

The primary objective of credit risk modelling is to answer three important questions:

1. What is the probability that the borrower will default?
2. If default occurs, how much loss will the lender suffer?
3. How much money is exposed at the time of default?

These questions are answered using three major components:

- Probability of Default (PD)
- Loss Given Default (LGD)

- Exposure at Default (EAD)

Credit risk modelling plays a central role in modern banking because it improves loan approval decisions, supports pricing of loans according to risk, helps maintain capital adequacy under Basel norms, reduces non-performing assets (NPAs), and strengthens overall financial stability.

Probability of Default (PD): Meaning and Concept

Probability of Default (PD) is the likelihood or chance that a borrower will fail to meet debt obligations over a specified period, generally one year. It is one of the most fundamental concepts in credit risk management because it measures the borrower's creditworthiness.

For example, if out of 100 borrowers with similar characteristics, 5 are expected to default within one year, then the PD is 5%.

PD can be mathematically expressed as:

PD = Number of Borrowers Who Default / Total Number of Borrowers

PD is usually represented as a percentage. A borrower with a higher PD is considered riskier, while a borrower with lower PD is considered more creditworthy.

Banks estimate PD by analyzing historical repayment patterns, financial statements, income stability, repayment behavior, and macroeconomic conditions.

Factors Influencing Probability of Default

1. Borrower-Specific Factors

The borrower's financial profile has a major impact on PD. Income stability, employment history, debt burden, previous repayment record, and credit score are key indicators. A salaried employee with stable income and strong repayment history generally has lower PD than someone with irregular income or prior defaults.

2. Financial Strength of Corporate Borrowers

For companies, profitability, leverage ratio, cash flow position, debt-equity ratio, and liquidity determine default probability. A company with strong profits and low debt has lower PD.

3. Macroeconomic Conditions

Economic slowdown, inflation, unemployment, and rising interest rates increase PD because borrowers may face financial stress. During recessionary periods, default rates typically rise.

4. Industry Risk

Certain sectors are more vulnerable to business cycles. For example, startups or cyclical industries may have higher PD than stable utility sectors.

Importance of PD in Financial Institutions

Financial institutions use PD extensively in assessing borrower risk. It forms the basis for:

- Loan approval or rejection
- Credit scoring
- Interest rate determination
- Credit rating assignment
- Regulatory capital calculation

For example, if two borrowers apply for loans, one with PD of 2% and another with PD of 10%, the bank may charge a higher interest rate to the second borrower or reject the application altogether.

Thus, PD helps lenders distinguish between low-risk and high-risk borrowers.

Loss Given Default (LGD): Concept and Meaning

Loss Given Default (LGD) measures the proportion of total exposure that a lender loses if the borrower defaults, after considering recoveries from collateral, legal proceedings, or asset liquidation.

In other words, LGD answers the question:
"If the borrower defaults, how much money will actually be lost?"

The formula is:

LGD = (Exposure at Default – Recovery Value) / Exposure at Default

It can also be expressed as:

LGD = 1 – Recovery Rate

For example, if a bank lends ₹10 lakh and recovers ₹6 lakh through collateral after default, then:

LGD = (10 – 6)/10 = 40%

Thus, the lender loses 40% of exposure.

Factors Determining LGD and Recovery Rate

1. Collateral Security

Secured loans generally have lower LGD because banks can recover losses through pledged assets such as property, machinery, or securities.

2. Type and Seniority of Debt

Senior secured creditors are paid before unsecured creditors during bankruptcy, reducing LGD.

3. Legal and Bankruptcy Framework

Countries with efficient legal systems and strong creditor rights usually have better recovery rates.

4. Economic Conditions

In weak economic conditions, collateral values may decline, increasing LGD.

5. Nature of Borrower Assets

Liquid assets such as real estate or marketable securities improve recovery, whereas intangible assets may have low liquidation value.

Importance of LGD in Credit Risk Management

LGD is essential because two borrowers with the same PD may generate very different losses depending on collateral quality. It helps banks:

- Price secured and unsecured loans differently
- Estimate recovery potential
- Calculate expected losses
- Set loan loss provisions
- Meet Basel capital standards

For instance, mortgage loans often have lower LGD than unsecured personal loans because homes serve as collateral.

Exposure at Default (EAD): Concept and Meaning

Exposure at Default (EAD) refers to the total value that a lender is exposed to when a borrower defaults. It represents the outstanding amount at the point of default.

For term loans, EAD is generally equal to the unpaid loan balance. However, for revolving credit facilities like credit cards or overdrafts, borrowers may draw additional funds before default, making EAD more complex.

For example, if a borrower has a credit card limit of ₹2 lakh and has used ₹1.5 lakh at default, EAD is based on current plus likely future utilization.

Importance of EAD

EAD helps lenders estimate the size of financial exposure. It is especially critical for:

- Credit cards
- Corporate credit lines
- Overdrafts
- Trade finance

Accurate EAD estimation ensures realistic capital reserves and better risk measurement.

Combined Use of PD, LGD, and EAD

These three components together determine Expected Loss (EL), which is the average loss a financial institution expects from a borrower.

Expected Loss Formula:

EL = PD × LGD × EAD

For example:
If PD = 4%, LGD = 50%, and EAD = ₹20,00,000

Expected Loss = 0.04 × 0.50 × 20,00,000 = ₹40,000

This means the institution expects an average loss of ₹40,000 from this exposure.

Role of Credit Risk Modelling in Borrower Evaluation

Credit risk modelling allows institutions to:

- Evaluate borrower quality scientifically
- Segment customers by risk profile
- Reduce defaults
- Improve profitability
- Allocate capital efficiently
- Conduct stress testing
- Comply with Basel II and Basel III norms

By combining PD, LGD, and EAD, institutions can build robust models for retail banking, corporate lending, and portfolio risk management.

Short Note: Credit Risk Modelling

Credit Risk Modelling is a structured analytical framework used by financial institutions to assess the probability and financial impact of borrower default. It combines statistical techniques with financial indicators to estimate default risk and expected loss. Its major components are PD, LGD, and EAD. It enhances lending decisions, pricing, capital allocation, and regulatory compliance.

Short Note: Exposure at Default (EAD) Model

The EAD model estimates the amount a lender is exposed to when a borrower defaults. It is especially useful for products with uncertain utilization, such as credit cards and revolving credit lines. EAD improves credit risk estimation by measuring potential exposure more accurately.

Conclusion

Credit risk modelling has become a fundamental part of modern banking and finance. It transforms uncertain borrower behavior into measurable financial risk. Probability of Default (PD) measures the likelihood of default, Loss Given Default (LGD) measures the severity of loss after default, and Exposure at Default (EAD) measures the total amount at risk. Together, these components form the foundation of expected loss estimation and strategic credit management.

In an increasingly complex financial system, effective credit risk modelling not only protects lenders from losses but also supports sustainable lending, regulatory compliance, and economic stability. For students of finance, understanding PD, LGD, and EAD is essential for grasping how institutions evaluate and manage borrower risk scientifically.